

Title: AI Hybrid Based Plagiarism Detection System Creation

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Abstract :

The paper focuses on the growing problem of plagiarism in academic environments due to increased digital content access. Traditional plagiarism detection methods, such as manual checking and rule-based systems, are often inefficient and inaccurate.

To address this issue, the authors propose an AI-based hybrid plagiarism detection system that uses machine learning and Natural Language Processing (NLP). The system analyzes textual content, detects similarities, and identifies plagiarism with high accuracy and efficiency.

The approach enhances scalability, speed, and accuracy, making it suitable for modern academic needs.

Introduction :

Plagiarism is a major threat to academic integrity, involving the use of others' ideas or content without proper acknowledgment. With the rapid growth of digital resources, detecting plagiarism has become more complex.

Traditional methods fail to handle large-scale data and advanced plagiarism techniques such as paraphrasing. The paper introduces AI-powered systems that combine machine learning and NLP to improve detection accuracy and efficiency.

These systems can analyze text deeply and identify hidden patterns of plagiarism, ensuring better protection of academic work.

Identified Research Gap :

The authors highlight several limitations in existing systems:

- Dependence on manual or rule-based methods
- Low accuracy in detecting paraphrased content
- Poor scalability for large datasets
- Lack of real-time detection
- Limited ability to analyze semantic meaning

The proposed AI hybrid system addresses these gaps by using advanced algorithms and linguistic analysis techniques.

Methodology :

The system follows a structured AI-based approach:

- Data Collection:
Large datasets of academic documents are gathered.
- Preprocessing:
Tokenization
Stop-word removal

- Stemming
- Feature Extraction:
 - Lexical features (word frequency)
 - Syntactic features (sentence structure)
 - Semantic features (meaning analysis)
- Model Training:
 - Machine learning models such as:
 - Support Vector Machines (SVM)
 - Gradient Boosting
- Similarity Detection: Cosine similarity Levenshtein distance
- System Integration:
 - A user-friendly interface for real-time plagiarism detection.
- Output:
 - Reports showing similarity percentage, sources, and flagged sections.

Results :

The proposed AI hybrid system demonstrates:

- High accuracy and efficiency compared to traditional methods
- Ability to detect:
 - Direct copying
 - Paraphrasing
 - Citation manipulation
- Improved precision, recall, and F1-score
- Faster processing of large volumes of data
- Reliable performance across different academic domains

The system also reduces manual workload and provides real-time feedback.

Conclusion :

The paper concludes that AI hybrid plagiarism detection systems are a major advancement in ensuring academic integrity. By combining machine learning and NLP techniques, the system provides accurate, fast, and scalable plagiarism detection.

Although challenges such as bias and false positives remain, proper implementation and ethical considerations can overcome these issues. The system is highly beneficial for educational institutions, researchers, and students, and represents the future of plagiarism detection technology.